

BEN ROSS

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EDUCATION

University of Maryland, College Park — B.S. Computer Science, direct admit *Expected May 2029*
• FIRE (First-Year Innovation & Research Experience), invited program. National Merit Finalist · SAT 1570, single sitting · 99th percentile, National Cyber League Fall 2025.

EXPERIENCE

U.S. Army Research Laboratory (DEVCOM) — *Research Apprentice, AEOP* *Jun – Aug 2026*
• Built a global GNSS spoofing detection pipeline over position reports ships and aircraft already broadcast publicly, using the ADS-B Navigation Integrity Category as a degradation proxy and a CNN over AIS vessel tracks to flag kinematically implausible movement.
• Shipped a dashboard that DBSCAN-clusters low-integrity reports to estimate emitter locations, with time-series views of how affected regions shift. Presented three times, including the ARL Director's Student Research Development Program poster session.
• Stood up on-premises LLM inference so lab engineers could use agentic coding tools without data leaving the network: Muse Glimmer 30B on llama.cpp on an NVIDIA A100, Open WebUI for per-user auth and request logging, Docker and Caddy for reverse proxy and TLS, and an installer provisioning that endpoint onto engineer workstations with per-user keys and certificates.

Chalgen — *Sole Developer* *May 2025 – Mar 2026*
• Built an agentic harness in which an LLM writes, runs and verifies its own forensics challenges through tool calls (disk images, PCAPs, email, EXIF, encrypted PDFs, databases), taking success from 0% with chat prompting to first-try passes.
• Benchmarked Claude Sonnet 4.5, GPT-5.1 Codex and Gemini 3 Pro on success, time and cost; drafted a full storyline competition in 15 minutes, and built the framework Montgomery College ran its March 2026 competition on.

Sparrows — *Lead Developer & Outreach Manager* *2023 – Aug 2026*
• Built and ran a multi-tenant school platform on Next.js 14, TypeScript and MongoDB, serving 30+ schools and roughly 30,000 monthly active users as an official partner of the Montgomery County Board of Education.
• Shipped per-school routing and admin dashboards, NextAuth credential auth with bcrypt and JWT, English and Spanish i18n, a PWA, and AI schedule parsing on the Gemini API, deployed on AWS Amplify with CloudWatch monitoring.

PROJECTS

SMCS Kart — *drive-by-wire electric go-kart* · C++, Python, React · github.com/PHS-SMCS/gokart-dash *2025 – 2026*
• Pitched a fully electric kart (40 mph, 40 hp, 72 V) when the head of the magnet program asked two students for a way to restructure the robotics curriculum, then built it and taught the class from it.
• Designed the control stack around one motion authority: a Teensy 4.1 owns the state machine, pedals, watchdogs and throttle DAC, an ESP32 runs PID steer-by-wire over 1 Mbps CAN, and the Pi dashboard is advisory only, so it can crash without affecting driving.

Planet Falconia — *VR digital twin, with NIST* · ParaView, OpenCV, LiDAR, VR *2023 – 2026*
• Integrated VR into a rover mission on a physical simulated-planet terrain: a walkable digital twin built from LiDAR and photogrammetry scans in ParaView, an altimetry heightmap, and AprilTag rover tracking with OpenCV.
• First high school students to speak at the Johns Hopkins APL XR Symposium, 2025; wrote most of the SIGGRAPH Asia talk below.

Herald — *open-source personal agent* · Python, SQLite, MIT *2026 – present*
• A personal agent running entirely on the owner's machine: ingests mail, calendar, messages, location and health data into one SQLite ledger, then runs scheduled reasoning cycles against it, architected so conclusions are maintained rather than recomputed.
• Graded autonomy enforced structurally rather than by prompt: the write path has no send function, every non-read action is logged, and the surface collecting an approval cannot carry it out. Shipped with migrations and an extension system.

PUBLICATIONS

S. Su, W. Sherman, J. Terrill, Z. Kingman, E. Ferrara, **B. Ross**, et al. “Planet Falconia: Mission Possible Using Immersive Visualization to Enable Digital Twin of Remote Sensing Data.” *SIGGRAPH Asia 2026 Educator’s Forum* (accepted).

LEADERSHIP

Captain & Instructor, Poolesville Computer Team — wrote the curriculum and lectured 40+ students twice weekly on cybersecurity and ML; entered seven CyberPatriot 18 teams, four reached national semifinals. *2024 – 2026*

SKILLS

Languages — Python, Java, C++, JavaScript/TypeScript, SQL, Bash

ML & data — TensorFlow, CUDA, OpenCV, llama.cpp, LLM fine-tuning, CNNs, clustering, anomaly detection, SQLite, MongoDB

Systems — Linux, Docker, AWS, Google Cloud, Cloudflare, Vercel, Next.js, React, Teensy, ESP32, Raspberry Pi, CAN bus